Testing and debugging functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Testing and debugging functions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reading structured data

Without opening the old lesson, explain one key idea from Reading structured data. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Return values and local state

Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

Complete the supported Testing and debugging functions trace and record the first step you can explain without help.

Use W2: Find a revealing case for Testing and debugging functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Testing and debugging functions explanation against the supplied facts before discussing it.

Use the worked case for Testing and debugging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Use boundary tests to reveal a function defect

Predict the **output**. If the function mistakenly uses x<=10, identify the first failing **assertion** and whether the final print happens. State the repair and which earlier tests must be rerun.

Do these cases prove all **input** types are handled?

```
def below_limit(x):
    return x < 10
assert below_limit(9) is True
assert below_limit(10) is False
assert below_limit(11) is False
print("three checks passed")
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Write expectations

Write expectations. Apply the supplied Testing and debugging functions facts and keep your reasoning.

Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Find a revealing case

Find a revealing case. Apply the supplied Testing and debugging functions facts and keep your reasoning.

A faulty function always returns 4. Which of your three cases pass and fail?

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair and retest

Repair and retest. Apply the supplied Testing and debugging functions facts and keep your reasoning.

A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a **regression** case.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Test a new function independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Test a new function independently. Apply the supplied Testing and debugging functions facts and keep your reasoning.

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Limit the conclusion

> Limit the conclusion. Apply the supplied Testing and debugging functions facts and keep your reasoning.

All three stock tests pass. Does that prove correct handling of negative counts?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Testing and debugging functions to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled testing and debugging functions **evidence** sheet to Python readiness portfolio. Show how you can design revealing tests for a function.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Testing and debugging functions

> Use a test that catches the boundary bug. A function should classify an integer as tiny exactly when it is less than three. Explain the labelled step before checking the changed condition.

Use a test that catches the **boundary** bug. A function should classify an integer as tiny exactly when it is less than three.

Python 3 intended function: return x<3. **Test case**s: (2,True),(3,False),(4,False). Candidate bug: return x<=3.

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Testing and debugging functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record contract

Strict less-than excludes the value 3.



2. Run boundary cases

The intended outputs are True, False, False for 2,3,4.



3. Locate the bug

Using <= passes 2 but fails the test at 3 before the later test.

CHANGE ONE THING

Remove the test at 3 and keep only 2 and 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Predict the three correct **output**s. Which test first catches <=, and why do tests at only two and four miss the bug?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use a test that catches the **boundary** bug. A function should classify an integer as tiny exactly when it is less than three.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Using <= passes 2 but fails the test at 3 before the later test.

1. Locate the bug

2. Record contract

3. Run **boundary** cases

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Locate the bug Using <= passes 2 but fails the test at 3 before the later test. A revealing test distinguishes a correct implementation from a plausible mistake.

Checking only values far from the **boundary** would miss this inclusive-versus-strict bug.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the test at 3 and keep only 2 and 4. Which conclusion is supported by the supplied facts?

1. The buggy <= function passes those remaining tests.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The buggy <= function passes

> those remaining tests. Both procedures agree away from the equality **boundary**.

Repair a missing building block

Review Reading structured data (python-bridge/11). Goal: **Process** a list of records consistently. Short practice: Circle the included rows, calculate the sum and mean, and state coverage.

Repeat after adding the fourth row with count four.. Return to this week and explain what you changed.

Review Return values and local state (python-bridge/9). Goal: Use a returned value in a later calculation. Short practice: Trace the two returned values and explain why first stays seven.

What happens if the caller tries to print result directly?. Return to this week and explain what you changed.

Review Loops and trace tables (python-bridge/6). Goal: Trace variable values before and after each loop iteration. Short practice: List the four visited values and their running totals.

Explain why 5 is absent and what prints for range(1,1).. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

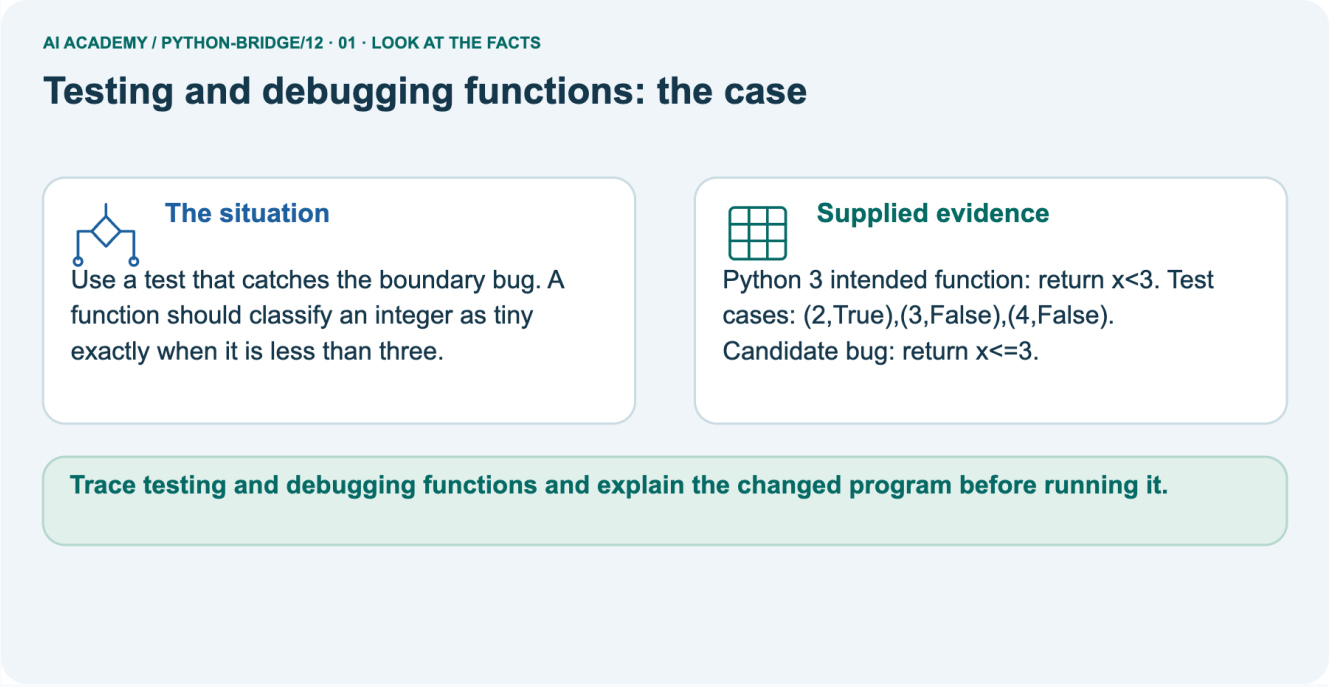
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Testing and debugging functions picture from its labels, then record a relevant limit.

01. Testing and debugging functions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

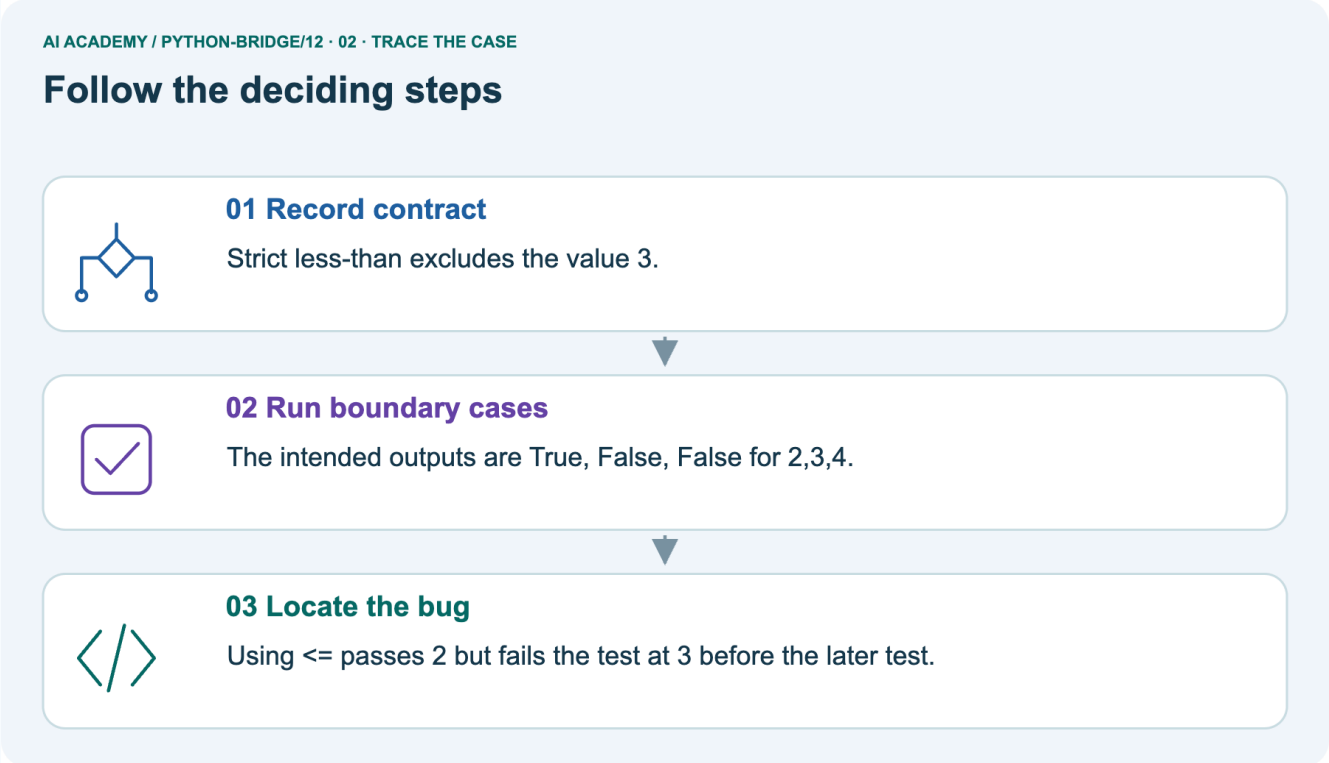
Use a test that catches the **boundary** bug. A function should classify an integer as tiny exactly when it is less than three. Supplied facts: Python 3 intended function: return x<3. **Test case**s: (2,True),(3,False),(4,False).

Candidate bug: return x<=3.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Record contract: Strict less-than excludes the value 3. → Run **boundary** cases: The intended **output**s are True, False, False for 2,3,4. → Locate the bug: Using <= passes 2 but fails the test at 3 before the later test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

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Follow the evidence

Step or case	Evidence	Meaning
Ordinary case	8 and 3	Expect 5
Full case	8 and 8	Expect 0
Unused case	8 and 0	Expect 8

Read each row from left to right and connect the evidence to the stated rule.

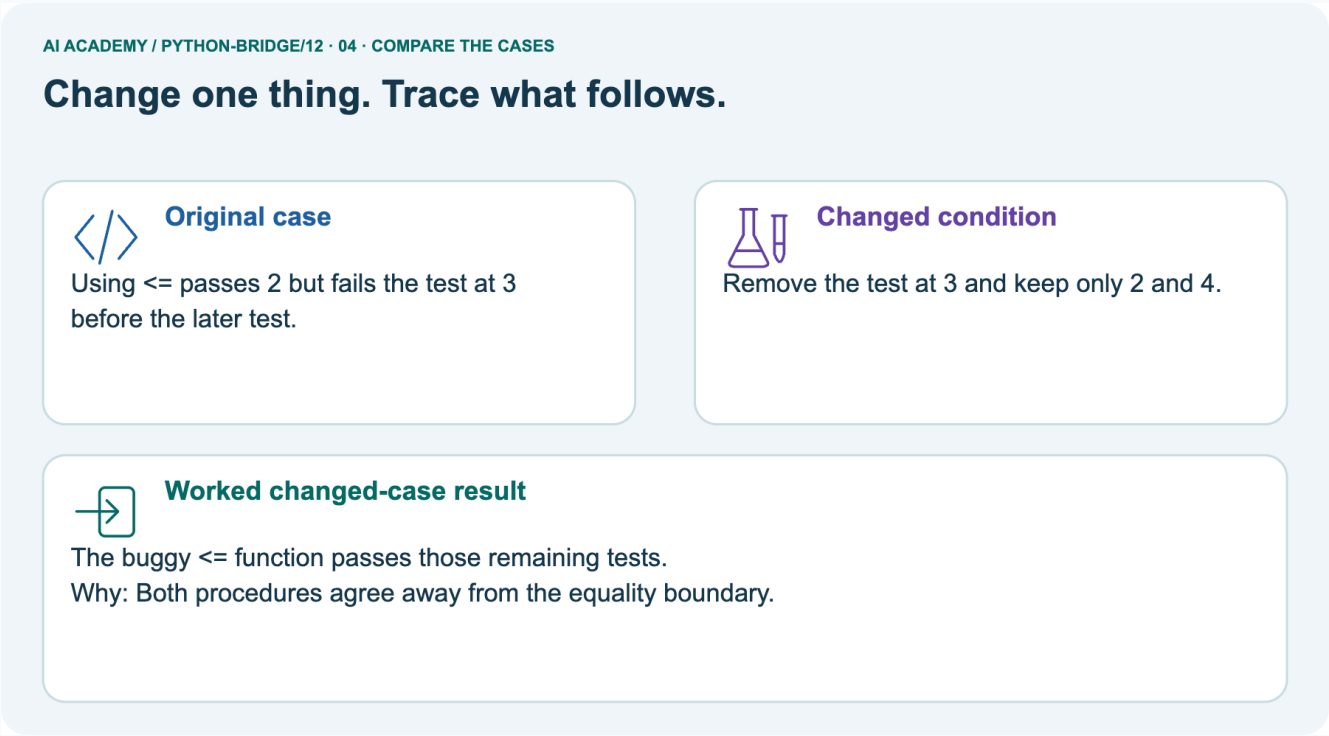
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Ordinary case | 8 and 3 | Expect 5. Full case | 8 and 8 | Expect 0. Unused case | 8 and 0 | Expect 8

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

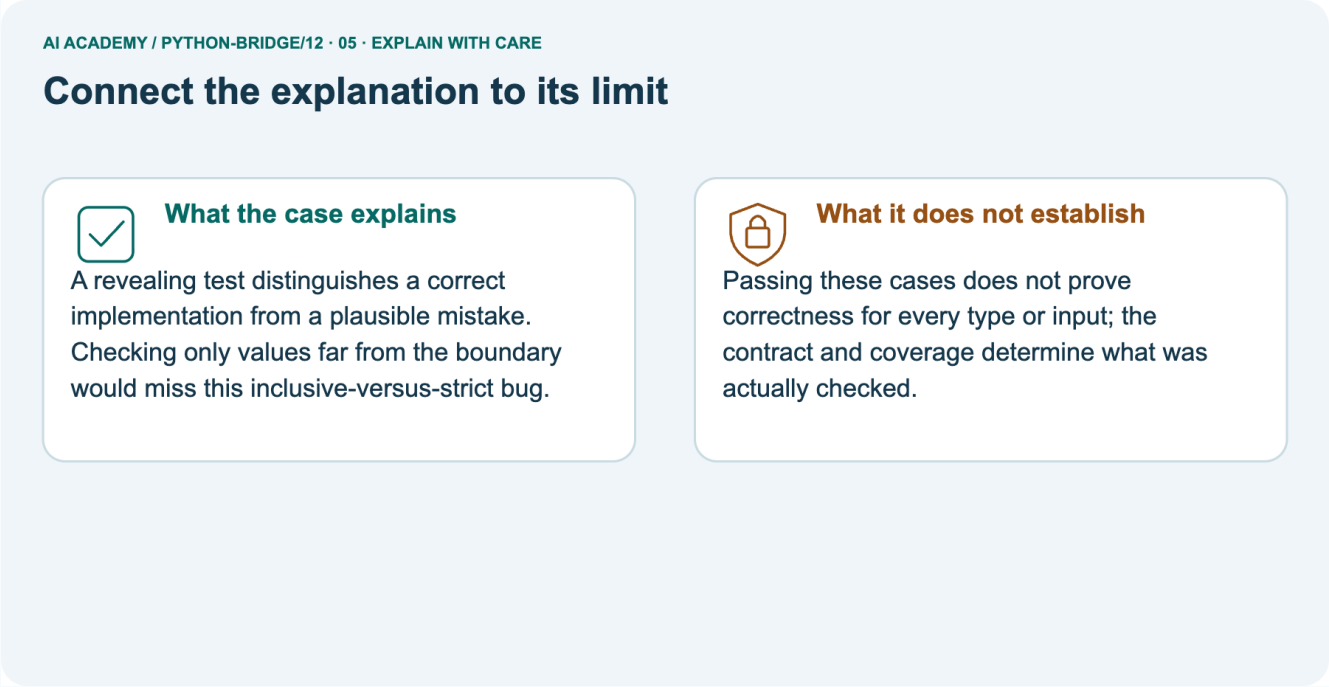
Original case: Using <= passes 2 but fails the test at 3 before the later test. Changed condition: Remove the test at 3 and keep only 2 and 4. Expected result: The buggy <= function passes those remaining tests.

Reason: Both procedures agree away from the equality **boundary**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

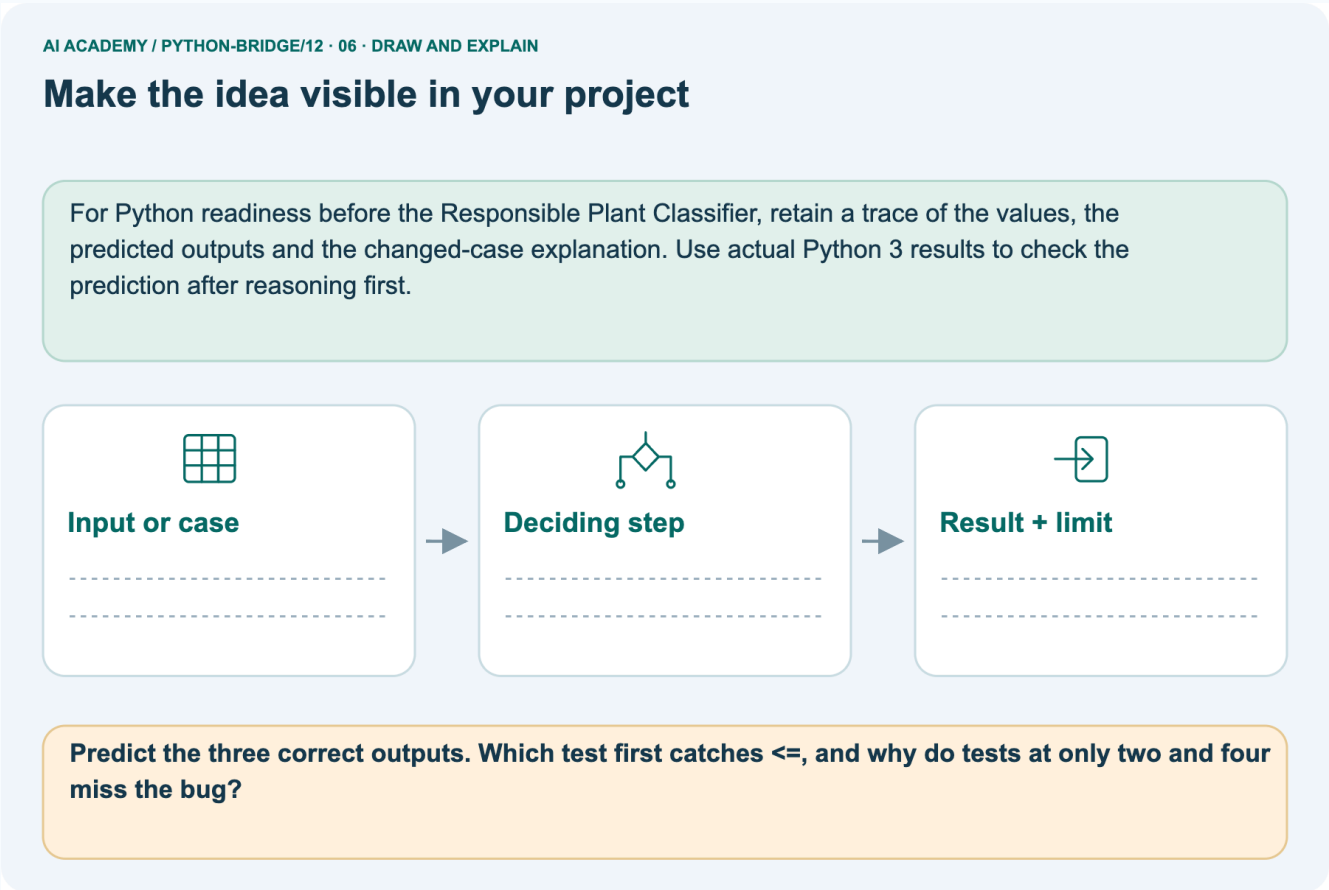
A revealing test distinguishes a correct implementation from a plausible mistake. Checking only values far from the **boundary** would miss this inclusive-versus-strict bug.

Limit: Passing these cases does not prove correctness for every type or **input**; the contract and coverage determine what was actually checked.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict the three correct **output**s. Which test first catches <=, and why do tests at only two and four miss the bug?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.